

# Moein Bagheri

Building AI that listens · Speech & Audio AI

[moeinbagheri.ai@gmail.com](mailto:moeinbagheri.ai@gmail.com) · [github.com/Moein-Bagheri](https://github.com/Moein-Bagheri) · Isfahan, Iran

## RESEARCH INTERESTS

Speech recognition for low-resource and code-switched languages (Persian–English) · Spoken language understanding · Audio and music representation learning

## EDUCATION

**B.Sc. in Computer Engineering, University of Isfahan**

Sep 2022 – Expected 2027

- GPA **17.26/20** overall · **18.5/20** over the last three semesters · **18.5/20** in specialized courses
- Major track: Artificial Intelligence · Minor track: Information Technology
- Selected courses: Signals & Systems (20) · Language & Speech Processing (20) · Design of Algorithms (20) · Data Mining (19.75) · Probability & Statistics (19.25) · Applied Linear Algebra (18.87) · Deep Learning (18.5) · Information Retrieval (18.25) · Artificial Intelligence (18)
- Final-year project (in progress): Persian–English code-switched speech recognition · Supervisor: Dr. S. F. Noorbehbahani

**NODET (SAMPAD) High School, Yazd**

Diploma

*Admitted through the national entrance exam for Iran's schools for exceptional talents*

## SELECTED PROJECTS

**Speech Emotion Recognition**

PyTorch · librosa · 2026

- Built plain and residual 1D CNNs on MFCC features to classify four emotions in an IEMOCAP subset.
- Compared random and speaker-independent splits across six experiments: accuracy fell from 65.1% to 58.0% on unseen speakers, showing how random splits overstate generalization.
- Released the trained checkpoints and a command-line predictor for WAV files.

**Local Research Assistant (RAG)**

Ollama · FAISS · Streamlit · 2026

- Built a fully local question-answering system over four retrieval papers (REALM, DPR, RAG, FiD) that cites the PDF pages it draws on.
- Routed requests to six research tools, with a keyword fallback when the model returns an invalid tool call.

**Persian Information Retrieval (with S. Heydari)**

Python · Hazm · 2026

- Built a Persian search engine with Hazm preprocessing, an inverted index, Boolean retrieval and TF-IDF/BM25 ranking.

**N-gram Music Completion**

Python · 2026

- Modeled melodies as note-token sequences with unigram to trigram models, Lidstone smoothing tuned by leave-one-out cross-validation, and perplexity evaluation.

**Deep ECG Classification**

TensorFlow · 2025

- Trained GRU networks for five-class heartbeat classification (MIT-BIH) and normal/abnormal classification (PTBDB).

**DiaTech: Product Discovery (team of four)**

Startup Development course · 2025

- Extended the competitor analysis and ran user interviews for a type-2 diabetes self-care concept.

*More on GitHub: real-time hand gesture detection (YOLOv8), smile detection (MobileNetV2), neural style transfer (VGG19), text classification with word embeddings.*

EXPERIENCE

Computer Innovation Center, University of Isfahan

Aug 2025 – Present

Business Development Intern → Student Coordinator

- Conducted market research for two early-stage startup concepts: an online physiotherapy platform and a personalized children's storybook service.
- Contributed to the design of the center's innovation workspace ahead of its opening in January 2026.
- Took part in meetings with Hub Isfahan, Mobarakeh Steel's university partner, during its equipment partnership with the center.
- Entrusted with day-to-day responsibility for the workspace.

LEADERSHIP

- Hosted the Computer Engineering department's welcome ceremony for new students, at the faculty's invitation.
- Organized Mathcup, an inter-school mathematics competition, in high school.

SKILLS

|                 |                                                                            |
|-----------------|----------------------------------------------------------------------------|
| Programming     | Python, Java, C/C++, SQL, JavaScript                                       |
| Speech & ML     | PyTorch, TensorFlow/Keras, scikit-learn, librosa, OpenCV, Ultralytics YOLO |
| NLP & retrieval | Hazm, NLTK, FAISS, Ollama, retrieval-augmented generation                  |
| Tools           | Git, Streamlit, Figma                                                      |
| Languages       | Persian (native), English (fluent)                                         |

INTERESTS

Piano (classical repertoire) · Music production in FL Studio and Ableton. Wrote and released an album on Spotify ([listen to "Lost"](#)).